

PeptideAtlas, UniProt, and dbAMP

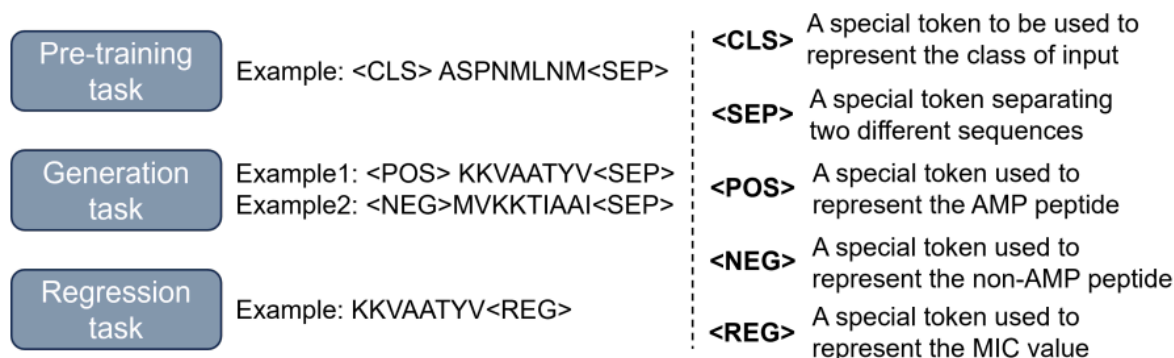
Over the past several days, I have spent countless hours modifying code and subsequently torturing my laptop's processor. Despite releasing several versions of the model, none have yet achieved the project's ultimate goal: generating novel scientific insights.

During these trials, I've curated far more data than I originally anticipated. I've collected over 600,000 unique peptide sequences from PeptideAtlas, along with more than 100,000 experimentally verified AMPs and non-AMPs. This dramatic increase in data should allow my current model to approach the standards set by the AMPCLGPT paper while reducing the risk of overfitting compared to the original 4,000-sequence dataset.

Contrastive Loss and MIC Prediction

The main challenge this past week has been implementing a loss function consistent with the AMPCLGPT paper, designed to effectively separate AMP from non-AMP sequences. Alongside this, the regression head for MIC prediction has pushed me to adopt a more sophisticated approach to sampling peptide data.

Previously, my model randomly batched groups of 128 amino acids, loosely concatenated from an AMP database. This approach proved ineffective, as it did not support the fixed-size training required for the regression head to learn MIC values. To address this, I explored several solutions and found that enforcing a maximum peptide length of 50 amino acids, combined with the introduction of new padding and delimiter tokens significantly strengthened the model's training process.



Redefining Scientific Insight

As I continue developing AMPForge2, I'm reminded of the initial passion that inspired me to create this model. My goal isn't to simply replicate existing projects, but to approach these specialized tasks in novel ways that generate new scientific insight.

Looking ahead, I plan not only to predict peptide MIC values but also to extend predictions to toxicity and stability. To achieve this, I intend to design a multi-task architecture that functions similarly to the current regression head while incorporating toxicity data from SATPdb and stability data from THPdb. This approach highlights the importance of integrating and curating data from diverse sources to build a more comprehensive training set.